Mathematics for Agricultural Science

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Mathematics for Agricultural Science (A14121)

Short Title:	Maths for Agricultural Science		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
Author	FLEONARD		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module introduces mathematical models and techniques used in agricultural and food processing applications, including problems in heat transfer applications, colony forming unit population destruction, separation using a centrifuge, viscosity, as well as elementary problems in blending and food processing using material balancing principles. The module also provides an introduction to the principles and practices of experimental design.

Programmes

MTHS-0050	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 4 / M

Indicative Content

- Material balancing: milk composition; cream and skimmed milk production; post processing potato crop yield; yeast growth; fertiliser blending
- Heat transfer: conduction through single and multiple layers of material; conduction through materials in parallel; convection in heat exchangers
- Thermal death times: logarithmic scales; D values; F values; Z values; single and multiple heat treatments
- Viscosity and separation: viscosity in a fluid; Reynolds number; streamline, transitional and turbulent flow; centrifuges; terminal velocity of separation particles
- Arrhenius equation: shelf-life determination of food products; calculations with experimental data
- Experimental design: principles of experimental design; randomisation, replication, blocking; experiment run order; ANOVA

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply the principles of material balancing to solve food processing problems.
2. Apply suitable heat transfer models of conduction and convection.
3. Apply models of centrifugal separation.
4. Determine the effect of a heat treatment on a colony forming unit population in a food product.
5. Calculate the shelf-life of a food product using the Arrhenius equation.
6. Apply experimental design principles to an agricultural problem,.
7. Apply models of the viscosity of fluids flowing in a cylindrical pipe.

Learning and Teaching Methods

- Lectures will be used to demonstrate numerical calculations.
- Students will do similar in-class exercises.
- Students will learn the basics of experimental design using suitable statistical software.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1,2,3,4,5,6,7

Assessment Criteria

<40%: Inability to identify or apply more than one appropriate mathematical model for agricultural science applications.

40%-49%: Able to identify and apply mathematical models for at least one agricultural science application.

50%-59%: Able to identify and apply mathematical models for a variety of agricultural science applications.

60%-69%: As above, and able to algebraically manipulate some models in agricultural science to explore the sensitivity of solutions.

70%-100%: As above, and able to algebraically manipulate a variety of models in agricultural science to explore the sensitivity of solutions.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	12	
Independent Learning	99	

Essential Material

- " NZIFST (Inc.) Unit operations in food processing." http://www.nzifst.org.nz/unitoperations/index.htm

Requested Resources

- Room Type: Computer Lab